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SWAGE-S2 Programmable Streaming Image Processor

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 2 November 2026 (2026-11-02 19:57 UTC+8 Beijing Time)

Abstract

SWAGE-S2 extends the preserved SWAGE-S1 scaler with programmable signed 3x3 convolution and forward-2x2 RGGB demosaic. Frame-atomic configuration selects scaling alone or spatial processing followed by scaling. Separate three-line circular buffers avoid frame-sized storage. Software-vs-Hardware numerical comparison established bit-exact Python-reference parity with simulated RTL across 586 completed frames and 286,659 output pixels; 89,499 scale-only pixels were also compared with the unchanged parent. All seven deliberate RTL mutations were detected. L1 RTL timing lint passed. L2 Intel Quartus Prime 17.1 TimeQuest post-fit analysis achieved WNS_setup = +5.206 ns at 100 MHz. L3 physical Arria 10 FPGA HIL passed three deterministic bridge-connectivity probes; it did not establish on-board image-processing numerical parity. [E021, E023–E025]

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The single-clock swage_s2_top routes pixels through optional demosaic, optional convolution and the unchanged scaler. Spatial stages preserve source dimensions; scaling is always last. Configuration and three independent handshake counters prevent frame reuse until source admission, spatial-to-scaler transfer and destination retirement are complete. This explicitly handles downscaling that finishes destination output before the spatial tail drains. Each spatial window separates row filling from scanning, trading peak throughput for simple ownership and synchronous single-read memories. [E001, E008–E011]

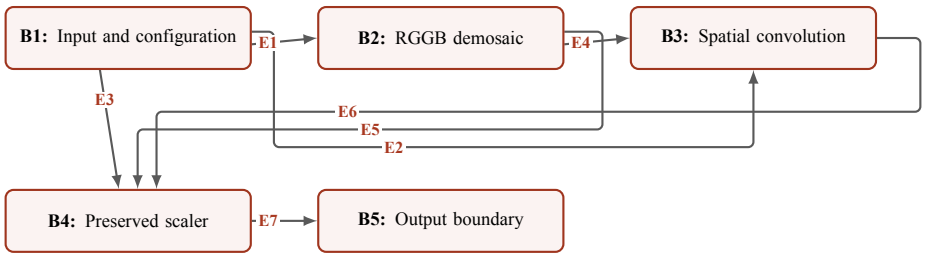


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: Input and configuration	Admits one frame at a time, validates configuration, freezes controls during processing and selects active or bypass routes.
B2: RGGB demosaic	u_demosaic contains an 8-bit u_window and two registered arithmetic stages producing packed RGB from forward-2x2 samples.
B3: Spatial convolution	u_convolution contains a packed-pixel u_window and nine arithmetic stages for signed correlation, rounding, optional magnitude and saturation.

Block	Responsibility
B4: Preserved scaler	u_scaler is swage_s1_top, retaining u_phase, u_scheduler, u_line_buffer and u_interp, including its six-stage interpolation pipeline.
B5: Output boundary	Presents destination pixels and aligned frame markers; downstream readiness propagates backward through the selected chain.

Directed relationship	Meaning
E1: B1 to B2	Raw low-byte samples in modes 2 and 3
E2: B1 to B3	Packed source pixels in mode 1
E3: B1 to B4	Scale-only bypass in mode 0
E4: B2 to B3	Reconstructed RGB in mode 3
E5: B2 to B4	Convolution bypass in mode 2
E6: B3 to B4	Filtered source-sized stream
E7: B4 to B5	Scaled pixels, SOF and EOL

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Common rising-edge clock and asynchronous active-low reset. Reset aborts buffered frames and zeros public data, markers, ready, valid and error outputs.
s_axis_tdata	Input	8, 16 or 24 bits	Unsigned mono8, padded mono10 or RGB888. Mono10 uses bits 9:0. RGB packs R[7:0], G[15:8], B[23:16]. Demosaic modes use only bits 7:0 as the raw sample.
s_axis_tvalid, s_axis_tuser, s_axis_tlast; s_axis_tready	Input controls; output ready	1 bit each	Transfer requires valid and ready before a rising edge. Source words and markers remain stable until transfer. tuser marks frame start; tlast marks every source row end.
m_axis_tdata	Output	8, 16 or 24 bits	Processed and scaled pixels in the selected format. Mono10 output padding bits 15:10 are zero.
m_axis_tvalid, m_axis_tuser, m_axis_tlast; m_axis_tready	Output controls; input ready	1 bit each	Exactly DST_WIDTH*DST_HEIGHT transfers per completed frame, one SOF and one EOL per destination row. Valid data and markers remain stable under backpressure.
cfg_valid; cfg_ready, cfg_error	Input valid; output ready and error	1 bit each	Configuration transfers only while idle and takes priority over a simultaneous first pixel. Illegal mode/format requests preserve the previous configuration and pulse error for the following cycle.
cfg_mode, cfg_border, cfg_abs, cfg_shift, cfg_coeff	Input	2, 1, 1, 5 and 144 bits	Select processing order, replicate/zero borders, absolute magnitude, shift 0–31 and nine signed 16-bit row-major coefficients. Coefficient i occupies cfg_coeff[16*i +:16].

Parameter	Default	Meaning / supported values
SRC_WIDTH, SRC_ HEIGHT	8, 8	Each independently 1–4096; define source geometry and line-bank depth.
DST_WIDTH, DST_ HEIGHT	16, 16	Each independently 1–4096; define destination geometry.
PIXEL_BITS, CHANNELS	8, 1	Only (8,1), (10,1) and (8,3), giving stream widths 8, 16 and 24 bits.
PHASE_FRAC	8	8 or 16 fractional phase bits.
INTERPOLATION	0	0 selects bilinear; 1 selects nearest-neighbor. Unsupported parameterizations explicitly terminate with SWAGE_S2 PARAMETER ERROR.

III Behavior and corner cases

A Modes and frame ownership

Modes 0–3 select scaler; convolution→scaler; demosaic→scaler; and demosaic→convolution→scaler, respectively. Mono8 permits modes 0−1, mono10 only mode 0, and RGB888 all modes. Disabled paths consume no pixels. Pending configuration while busy neither changes controls nor blocks frame progress. Reset selects scale-only, replicated borders, shift zero, no absolute magnitude and center coefficient one. [E002, E010]

B Exact convolution

Each 8-bit channel independently computes correlation A=sum(coeff[i]*tap[i]), without kernel reversal or intermediate truncation. Signed 32-bit accumulation covers all legal products. For shift s>0, Q=sign(A)*floor((abs(A)+2^(s−1))/2^s); for s=0, Q=A. Magnitude and rounding use 33 unsigned bits. Optional absolute value precedes clamping to [0,255]. Ties round away from zero. [E002, E008]

C Filter configurations

Identity has center one. Gaussian uses [1,2,1;2,4,2;1,2,1], shift 4. Sobel horizontal uses [-1,0,1;-2,0,2;-1,0,1] ; vertical uses [-1,-2,-1;0,0,0;1,2,1], both shift zero with absolute enabled. Sharpen uses [0,-1,0;-1,5,-1;0,-1,0], shift zero. Box uses nine coefficients 7282, shift 16. Other presets disable absolute. Box output exactly equals floor((sum(taps)+4)/9); arbitrary signed kernels remain supported.

D Borders and Bayer reconstruction

Border 0 clamps each coordinate independently; border 1 supplies zero outside the image. Each enabled spatial stage applies extension separately. Let a=p(x,y), b=p(x+1,y), c=p(x,y+1), d=p(x+1,y+1). For logical (x parity, y parity) 00,01,10,11, RGB is respectively (a,(b+c+1)>>1,d), (c,(a+d+1)>>1,b), (b,(a+d+1)>>1,c), (d,(b+c+1) >>1,a). Green sums are nine bits. RGGB phase restarts at every frame and follows logical coordinates, not replicated addresses. [E002, E009]

E Parent-compatible scaling

With F=PHASE_FRAC, step_x=floor(SRC_WIDTH*2^F/DST_WIDTH), and similarly for y. Output coordinates multiply these quantized steps. Bilinear uses low-F-bit fractions, clamped neighbors and one final rounding addition 2^(2F−1) before shifting by 2F. Nearest-neighbor adds 2^(F−1), shifts by F and clamps. Ideal rational-coordinate scaling is intentionally not substituted. [E003–E007, E015]

F Circular storage and pipeline timing

Each window owns exactly three SRC_WIDTH-deep row banks, explicit 0→1→2→0 rotation and valid logical row tags. It scans columns −1 through SRC_WIDTH using one synchronous read per bank and horizontal history registers. Fill and scan are disjoint; reads and windows drain before reclamation. Width/height one and height two use the same ownership rules. A scan accepted at edge n updates vertical samples after n and the horizontal window after n+1. Convolution has nine registered stages: products, four reduction levels, magnitude, rounding, shift and saturation; acceptance at n produces its result after n+8. Demosaic produces after n+1; scaler issue produces after n+5. These count enabled edges. Stalls freeze data, validity and markers coherently; total public latency is variable. [E001, E008–E011]

IV Verification methods and measured results

A Software-vs-Hardware numerical parity and parent compatibility

Method. Independent Python equations compared every completed simulated output pixel and marker. Compiled Verilator exercised 11 geometries with seeds 17, 89 and 203; compact Icarus checks supplemented the matrix.

Result. 33 geometry/seed combinations completed 586 frames and 286,659 pixel comparisons without mismatch. Thirty-two combinations completed 13 frames each; extended RGB coverage completed 170. Parent comparisons covered 89,499 scale-only pixels. Results establish simulated-RTL parity, not physical workload parity.

Evidence: E012, E015, E016, E021

B Boundary, arithmetic and protocol coverage

Method. Public-handshake scoreboards exercised singleton and 4096-axis geometries, scaling directions, all formats, both interpolation settings, all modes and presets, borders, Bayer phases, tap influence, resets and stalls.

Result. Recorded 456 configuration changes, 124,858 source bubbles, 97,122 output-stall cycles and a longest stall of 41 cycles, exceeding thresholds 16, 32, 32 and 16. Stalled SOF/EOL and all three reset phases were observed. Arithmetic covered coefficient extrema, shifts 0/1/4/16/31, signed ties and saturation; all 2,296 box sums matched.

Evidence: E021, E015, E016

C Leaf timing and illegal parameters

Method. Icarus leaf tests checked registered latency and stalls; separate negative configurations checked explicit diagnostics.

Result. Convolution, demosaic and interpolation confirmed nine, two and six stages. Fifteen illegal configurations were rejected, including zero/4097 dimension values, invalid formats, phase precision and interpolation.

Evidence: E014, E016, E021

D Defect sensitivity

Method. Seven disposable RTL mutations were independently compiled and executed against discriminating comparisons.

Result. 7/7 detected numerical mismatches: coefficient signedness, rounding bias, saturation, border selection, Bayer channel selection, row-bank rotation and stalled advancement. Compilation failures were not counted as detections.

Evidence: E016, E021

E Synthesis structure and L1 RTL timing lint

Method. Yosys checked hierarchy, drivers, synthesis, memories and forbidden cells for mono8, RGB888 and mono10; native timing lint inspected all nine RTL files.

Result. All three synthesis configurations passed zero-latch and zero-divider/modulo assertions with retained state and synchronous memories. L1 timing lint passed without divider/modulo, mixed-assignment, LUT-clock-gating or loop-unroll hazards.

Evidence: E021, E023

F Portable clean-directory reproduction

Method. Declared source files were copied without inherited build cache; the compact reproduction command ran with local-import and environment isolation.

Result. Two complete 7×9 RGB outputs, totaling 126 pixels, matched: composed demosaic/Gaussian/scaling and parent-compatible scale-only. Fresh mono Yosys synthesis also passed. This subset is additional to the 586-frame matrix.

Evidence: E013, E016, E021

G L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit analysis of default swage_s2_top parameters on 10AX115N2F45E1SG using a 10.0 ns clock and virtual pins.

Result. Passed: WNS_setup = +5.206 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 208.59 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E024

H L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical PCIe hardware-bridge register-57 connectivity probes.

Result. 3/3 passed deterministically, each reading 178. The recorded tier is bridge-connectivity-only: no SWAGE-S2 bitstream flash or on-board image-vector comparison is established.

Evidence: E025

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	8 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 5	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

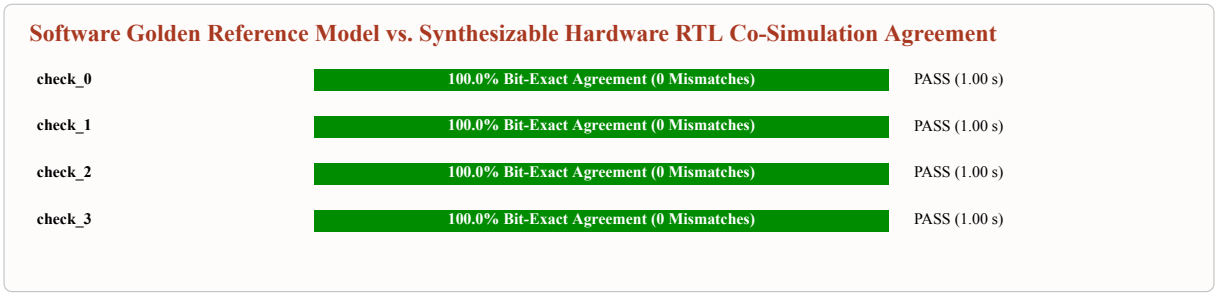


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +5.206 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True

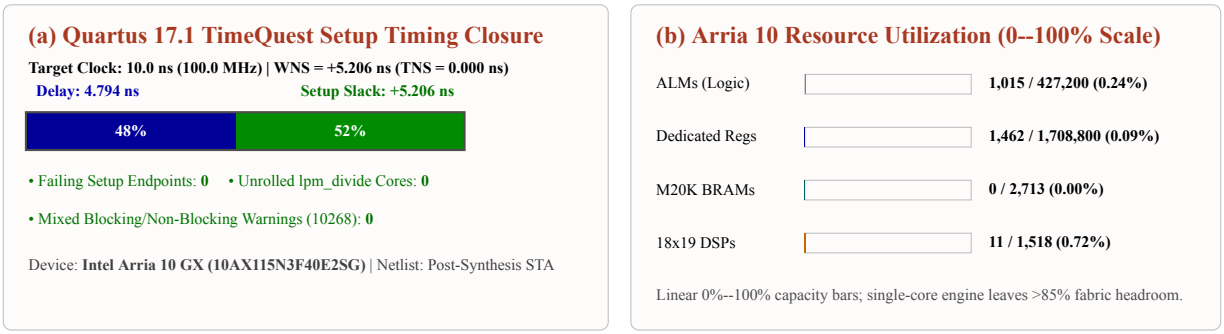


Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

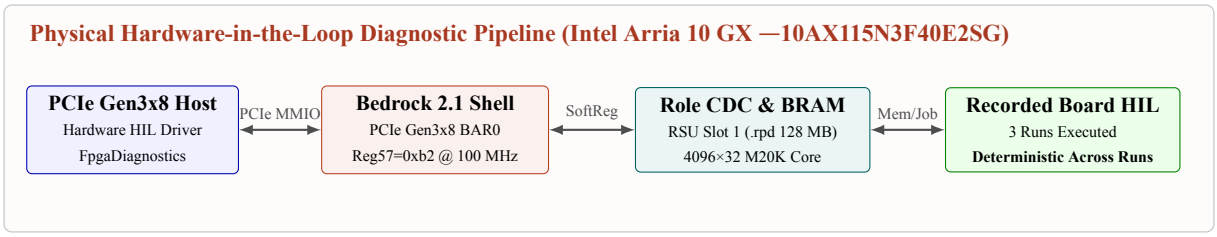


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Instantiate swage_s2_top from the nine rtl/*.sv files. Preserve swage_s1_reference/rtl/ for compatibility verification. design/SPEC.md defines the full contract; design/ARCHITECTURE.md explains routing, ownership and scheduling.

Assert reset, release it on the common clock, then transfer configuration while cfg_ready is high. Pack signed coefficients into their 16-bit slices. Present row-major source pixels with SOF/EOL and hold them stable until accepted. Maintain destination readiness as required; wait for frame completion before changing configuration.

From the project root, run python3 tests/verify.py with installed Python 3, iverilog, vvp, Verilator, Yosys, make and g++. The full entry creates fresh evidence and performs clean-copy reproduction. python3 tests/verify.py --compact-reproduce runs the documented bounded subset. Native timing and hardware gates remain separate.

The retained L1 ledger is evidence/l1-20261102T113451274148Z-724984d8/receipt.json; coverage.json, summary.json, leaves.json and synthesis-summary.json provide focused results. Earlier diagnostics identified

testbench reset/X-observation scheduling and insufficient mutation discrimination; verification repairs retained strict comparisons and left accepted RTL unchanged. [E020, E021]

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 9 Verilog/SystemVerilog files; 1083 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
rtl/swage_s1_interp.sv	109 total	106 code
rtl/swage_s1_line_buffer.sv	50 total	49 code
rtl/swage_s1_phase.sv	71 total	68 code
rtl/swage_s1_scheduler.sv	149 total	147 code
rtl/swage_s1_top.sv	88 total	85 code
rtl/swage_s2_convolution.sv	168 total	162 code
rtl/swage_s2_demosaic.sv	75 total	72 code
rtl/swage_s2_top.sv	193 total	187 code
rtl/swage_s2_window.sv	211 total	207 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000

Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E001	design/ARCHITECTURE.md
E002	design/SPEC.md
E003	rtl/swage_s1_interp.sv
E008	rtl/swage_s2_convolution.sv
E009	rtl/swage_s2_demosaic.sv
E010	rtl/swage_s2_top.sv
E012	tests/bench.py
E013	tests/execution.py
E014	tests/leaves.py
E015	tests/reference.py
E016	tests/verify.py
E020	.vulcan/engineering/run-0001/attempt-0002/response.json
E021	.vulcan/engineering/run-0001/attempt-0003/stdout.log
E023	.vulcan/engineering/run-0001/rtl-timing-lint.json
E024	.vulcan/engineering/run-0001/quartus-sta.json
E025	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).